

ELLIPTIC CURVES ASSOCIATED TO ABELIAN THREEFOLDS WITH IMAGINARY MULTIPLICATION

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MAIN RESULT

M imaginary quadratic field.

A an abelian threefold defined over a number field K .

A has imaginary multiplication (IM) by (M, ι) , if $\iota: M \hookrightarrow \text{End}^0(A_K) := \text{End}(A_K) \otimes \mathbb{Q}$, is an algebra embedding AND the e-values of $\iota(\alpha)$ acting on $\Omega(A)$ are $\sigma(\alpha), \sigma(\alpha), \overline{\sigma(\alpha)}$ where $\sigma: M \hookrightarrow \overline{M}$ fixed embedding.

A has potential imaginary multiplication (PIM) by (M, ι) , if A_{KM} has IM by (M, ι) .

Example. If $\text{End}^0(A_{\overline{K}}) \cong M$, then A has PIM. If $A \sim E \times B$ where B is an abelian surfaces such that $\text{End}(B) \not\cong M$, then A does not have imaginary multiplication by M .

C nice curve, $\tau \in \text{Aut}(C)$, $J = \text{Jac}(C)$. Does J have PIM by $(\mathbb{Q}[\tau], \iota)$ where $\iota: \mathbb{Q}[\tau] \hookrightarrow \text{End}^0(J)$? (Note $\mathbb{Q}[\tau]$ is an algebra by definition, but might not be a field!)

C	$\deg f$	$\tau(x, y)$	Basis of $\Omega(C)$	e-values	PIM
$y^3 = f(x)$	4	$(x, \zeta_3 y)$	$\frac{dx}{y^2}, \frac{x dx}{y^2}, \frac{dx}{y}$	$\zeta_3, \zeta_3, \zeta_3$	✓
$y^2 = f(x)$	7, $f(-x) = -f(x)$	$(-x, iy)$	$\frac{dx}{y}, \frac{x dx}{y}, \frac{x^2 dx}{y}$	$i, -i, i$	✓
$y^2 = f(x^4)$		(ix, y)		$i, -1, -i$	X
$y^4 = f(x)$	3	(x, iy)	$\frac{dx}{y^3}, \frac{x dx}{y^3}, \frac{dx}{y^2}$	$i, i, -1$	X

Last two cases $(\tau + 1)\mathbb{Q}[\tau] \cong \mathbb{Q}(i)$, but action of $\tau + 1$ on $\Omega(J)$ has rank 2, so not PIM either.

Theorem (Fité, G. '25). *Let A/K be an abelian threefold with PIM by (M, ι) , then $\text{Cl}(M) \leq [KM: M]$.*

λ -adic representations. A/K ab. var. dim 3 with IM by (M, ι) .

ℓ prime number.

$V_\ell(A)$ is a free $M \otimes \mathbb{Q}_\ell$ -module of rank 3.

The decomposition $M \otimes \mathbb{Q}_\ell \cong \prod_{\lambda|\ell} M_\lambda$ induces a decompositions of $G_K := \text{Gal}(\overline{K}/K)$ -modules

$$V_\ell(A) \cong \prod_{\lambda|\ell} V_\lambda(A),$$

with each $V_\lambda(A)$ a M_λ v.s. of dim 3.

Theorem (Main Theorem: Fité, G. '25). *Let A/K be an abelian threefold with PIM by (M, ι) . If $KM \neq K$, assume there is an embedding $\sigma: K \hookrightarrow \mathbb{C}$ such that either $\sigma(K) \subseteq \mathbb{R}$ or $j(\mathcal{O}_M) \in \sigma(K)$.*

Then there exists an elliptic curve E/K with potential CM by M such that, for every prime λ of M over a rational prime ℓ , the following is satisfied:

i) If $KM = K$, then

$$(\wedge^3 V_\lambda(A) \oplus \wedge^3 V_{\overline{\lambda}}(A))(-1) \simeq V_\ell(E) \otimes_{\mathbb{Q}_\ell} M_\lambda$$

as $M_\lambda[G_K]$ -modules.

ii) If $KM \neq K$, then

$$\text{Ind}_{KM}^K (\wedge^3 V_\lambda(A_{KM}))(-1) \simeq V_\ell(E) \otimes_{\mathbb{Q}_\ell} M_\lambda$$

as $M_\lambda[G_K]$ -modules.

proof overview: Case i)

$$\{\text{ECs } /K \text{ with CM by } M\} \longleftrightarrow \{\text{Algebraic Hecke character of } K \text{ with } \begin{matrix} \text{values in } M \text{ and } \infty\text{-type } (1,0) + \dots \end{matrix}\}$$

“ $\rightarrow$ ” Shimura-Taniyama and “ $\leftarrow$ ” Casselman. The above both have natural λ -adic reps of G_K attached to them and the correspondence preserves these.

$$\begin{array}{ccccc} A/K \text{ abelian threefold} & & (\wedge^3 V_\lambda(A))_\lambda \text{ compatible} & & \text{Alg. Hecke character of } K \text{ with} \\ \text{with IM by } (M, \iota) & \rightarrow & \text{system of reps of } G_K & \rightarrow & \text{values in } M \text{ and } \infty\text{-type } (2, 1) \end{array}$$

Twisting by the norm⁻¹ gives us an algebraic Hecke character whose ∞ -type is a $(1, 0)$. From here we obtain the CM elliptic curve in the IM case.

For case *ii*), we base change to MK and then use descent to prove the result over K . $\square$

FINAL REMARKS

Proof of i) works more generally, for “shifted CM types” e.g. A/K , $\dim A = 6$, $\iota: \mathbb{Q}(\zeta_5) \hookrightarrow \text{End}^0(A)$ signature $(2, 1, 2, 1)$ where $(1, 0, 1, 0)$ is a CM type for $\mathbb{Q}(\zeta_5)$. Proof of ii) is harder, but possible in some cases.